

Metagenomic Profiling of the Ocular Surface Microbiome in Patients After Allogeneic Hematopoietic Stem Cell Transplantation



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• **PURPOSE:** To investigate the characteristics of the ocular surface microbiome in patients after allogeneic hematopoietic stem cell transplantation (allo-HSCT) and the associations between the microbial dysbiosis and chronic ocular graft-versus-host disease (oGVHD).

• **DESIGN:** Prospective cohort study.

• **METHODS:** Ocular surface samples from 48 healthy subjects and 76 patients after allo-HSCT, including 50 patients with chronic oGVHD and 26 patients without oGVHD, were collected. Species-level composition of the ocular surface microbiome was surveyed via metagenomic shotgun sequencing. OGVHD was diagnosed and graded according to the International Chronic Ocular GVHD Consensus Group criteria.

• **RESULTS:** The α -diversity of the microbiota was significantly decreased in patients after allo-HSCT. Nevertheless, we detected more types of viral species in the allo-HSCT group than the healthy group, especially anelloviruses. The mismatch of donor-recipient sex was only negatively associated with the α -diversity in male but not female recipients. Moreover, the microbiome of patients with oGVHD was distinct from patients without oGVHD. *Gordonia bronchialis* and *Pseudomonas parafulva* were enriched in patients with oGVHD and positively associated with International Chronic Ocular GVHD score.

• **CONCLUSIONS:** This study suggests that the ocular surface microbiome after allo-HSCT is characterized by a loss of diversity. Furthermore, the microbial dysbiosis at the ocular surface is associated with the status and severity of chronic oGVHD. These results lay the groundwork for future investigations of the potential microbial mecha-

nism for oGVHD. (Am J Ophthalmol 2022;242: 144–155. © 2022 Elsevier Inc. All rights reserved.)

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is the strongest curative treatment for patients with hematologic malignancies. Unfortunately, graft-versus-host disease (GVHD), a common complication after allo-HSCT, is a leading cause of mortality.¹ Ocular involvement, which can be a precursor to systemic GVHD and a predictor of poor prognosis including mortality, occurs in 60% to 90% of patients with chronic GVHD and significantly impairs their visual function and quality of life.^{2,3} Given the increasing number of transplant recipients, there will be an increasing population of patients with chronic GVHD involving the eyes. Chronic ocular GVHD (oGVHD) involves multiple sites at the ocular surface, including the lacrimal gland, meibomian gland, conjunctiva, and cornea.⁴ The most common ocular manifestation of chronic oGVHD is refractory dry eye disease. Blindness caused by cornea ulcer and perforation may occur in severe cases. The pathogenesis of oGVHD is poorly understood, and the overall therapeutic outcome is unsatisfactory.

The human microbiome plays an essential role in immunoregulation and contributes to immune homeostasis and disease pathogenesis. Microbial dysbiosis may facilitate the development or exacerbation of multiple immune-mediated diseases.⁵ Extensive investigations in the gut microbiome have demonstrated that microbial dysbiosis is involved in allo-HSCT and linked to the risk of GVHD.^{6,7} Recent studies based on culture and 16S rRNA sequencing methods provided initial evidence that the ocular surface microbiota may play a role in oGVHD.^{8–10} However, the composition of the ocular surface microbiome in patients after allo-HSCT and the relationship between the microbiome and oGVHD are far from being fully characterized. In this study, we surveyed the ocular surface microbiome using metagenomic shotgun sequencing for allo-HSCT recipients with and without chronic oGVHD and healthy controls. Our study may shed light on a better understanding of the pathogenesis of oGVHD and the development of novel therapeutic approaches in the future.

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METHODS

• **PARTICIPANT RECRUITMENT:** This study was conducted in accordance with the tenets of the Helsinki Declaration of Human Studies. All procedures were performed in compliance with the protocol (#2019KYPJ135) approved by the Ethics Committee of Zhongshan Ophthalmic Center, Sun Yat-sen University. Participants were recruited at Zhongshan Ophthalmic Centre from January 2019 through December 2020. Written informed consent was obtained from all participants by agreement document. Patients who underwent allo-HSCT and came at our center for routine ocular surface check-up visits were screened for enrollment. Allo-HSCT recipients without oGVHD (non-oGVHD), patients with untreated oGVHD, and healthy controls were recruited.

Exclusion criteria were: (1) any ocular surface diseases other than oGVHD, such as infection, blepharitis, allergic conjunctivitis, Stevens-Johnson syndrome, and Sjögren's syndrome; (2) topical use of eyedrops or ointment within the past 3 months; (3) a history of ocular surgery or trauma within the past 6 months; (4) the presence of any other systemic diseases including active uncontrolled GVHD in other organs within the past 3 months; and (5) contact lens users or smokers.

The chronic ocular GVHD was diagnosed according to the criteria proposed by the International Chronic Ocular GVHD (ICO) Consensus Group.¹¹ The parameters for diagnostic include the Ocular Surface Disease Index (OSDI), Schirmer's test score, corneal fluorescein staining (CFS) score, and conjunctival hyperemia. Demographic information, comprehensive medical and ophthalmologic history, primary disease, donor/recipient sex, and radiotherapy/chemotherapy history were recorded for all the participants.

• **OPHTHALMIC EXAMINATION:** The severity of chronic oGVHD was graded according to the ICO diagnostic criteria.¹¹ The ICO score is based on an aggregate of scores for Schirmer's test, OSDI, CFS score, and conjunctival hyperemia.¹¹ The total score (points) and severity classification are as follows: 0-4, none; 5-8, mild/moderate; 9-11, severe.¹¹ CFS ranged from 0 to 3 points (grade 0 = no staining, grade 1 = minimal staining, grade 2 = mild/moderate staining, grade 3 = severe staining) according to the International Consensus Criteria on chronic ocular GVHD Group.¹¹ Schirmer's test was performed without anesthesia using sterile strips (Tear Flo; Sigma Pharmaceuticals), and the length of paper wetting was measured after 5 minutes.¹² The symptoms were scored with the OSDI questionnaire. The shape of the palpebral margin was observed under a slitlamp and checked for congestion of the palpebral margin, blockage of a meibomian gland opening, irregular shape of the palpebral margin, and backward displacement of a meibomian gland

orifice opening.^{13,14} The degree of meibomian gland loss (MGL) was examined using Keratograph 5M (Oculus).¹⁵ Both eyes of the participants were examined and tested for all the clinical parameters. The eye with a higher ICO score was selected for analysis. The right eye was designated as the study eye if the ICO scores were equal between the 2 eyes.

• **SAMPLE COLLECTION:** The conjunctival impression samples were collected and stored as follows. The eyes were topically anesthetized with 1 to 2 drops of Alcaine Eye Drop (Alcon). A sterile semicircle MF-Millipore™ Membrane Filter (MF) (REF: HAWP01300, 0.45 µm in diameter; Merck Millipore) was placed on the lower bulbar conjunctiva for 10 seconds. The membrane was then immediately placed in a sterile tube with 300 µL of Tissue and Cell Lysis Solution (Epicentre) and stored at -80 °C.

• **DNA ISOLATION AND METAGENOMIC SHOTGUN SEQUENCING:** DNA was extracted from each conjunctival sample using the MasterPure Complete DNA and RNA Purification Kit (Epicentre) according to the manufacturer's instructions. A total of 100 ng DNA per sample was sonicated into 300 to 400 bp fragments using Bioruptor (Diagenode). Sequencing libraries were prepared using the VAHTS Universal DNA Library Prep Kit for Illumina (Vazyme) and quantified by quantitative polymerase chain reaction (qPCR) using the KAPA SYBR FAST qPCR Kit (Kapa Biosystems). Paired-end 2 × 150-bp sequencing was performed on a NovaSeq 6000 instrument (Illumina). Unused membranes that did not contact the conjunctiva were used as negative blank controls. A total of 18 negative blank controls, which had reagents from DNA extraction through sequencing, were processed along with the samples.

• **TAXONOMIC AND FUNCTIONAL PROFILING:** The sequencing reads were first quality filtered using Trimmomatic¹⁶ v0.36 and PRINSEQ¹⁷ v0.20.4. Human reads were removed using KneadData v0.6.1 (<https://huttenhower.sph.harvard.edu/kneaddata>). High-quality nonhuman reads were mapped against a custom database using Kraken2¹⁸ v2.0.9. A total of 29,943 complete microbial genomes were downloaded on May 3, 2020, of which 19,362 were bacterial, 368 were archaeal, 9346 were viral, and 867 were fungal. The complete bacterial, archaeal, and viral genomes were downloaded from the RefSeq database using the download-library option of kraken2-build. The complete fungal genomes were manually downloaded from the GenBank database. The results of taxonomic classification were filtered using a confidence score of 0.20. The species profiles were decontaminated (see below). The reads derived from noncontaminant species were used for functional analysis. Functional annotation was performed using eggNOG-mapper¹⁹ v2.0.0 based on precomputed eggnog²⁰ v5.0.

TABLE 1. Demographic Characteristics for the Study Population

	oGVHD	Non-oGVHD	Healthy	P Value
Cases, n	50	26	48	
Sex (male/female)	28/22	16/10	23/25	.332 (healthy vs non-oGVHD) ^a .807 (oGVHD vs non-oGVHD) ^a .544 (oGVHD vs healthy) ^a
Age, mean (SD)	36.1 (11.3)	31.9 (11.6)	27.9 (4.0)	.097 (healthy vs non-oGVHD) ^b .139 (oGVHD vs non-oGVHD) ^b <.001 (oGVHD vs healthy) ^b
Donor sex (male/female)	29/21	14/12	—	.809 (oGVHD vs non-oGVHD)
Donor-recipient sex combination, n (%)			—	.376 (oGVHD vs non-oGVHD)
Male to male	16 (32.0)	6 (23.1)	—	
Male to female	13 (26.0)	8 (30.8)	—	
Female to female	9 (18.0)	2 (7.69)	—	
Female to male	12 (24.0)	10 (38.5)	—	
Level of matching, n (%)			—	.002 (oGVHD vs non-oGVHD)
Matched related donors	42 (84.0)	13 (50.0)	—	
Matched unrelated donors	8 (16.0)	13 (50.0)	—	
Post allo-HSCT days, median (IQR)	80.5 (23, 197)	181 (32, 444)	—	.301 (oGVHD vs non-oGVHD) ^c

allo-HSCT = allogeneic hematopoietic stem cell transplantation, IQR = interquartile range, oGVHD = ocular graft-versus-host disease.

^aFisher's exact *t*-test.

^bUnpaired *t*-test.

^cMann-Whitney *U*-test; the Pearson χ^2 test was used unless otherwise mentioned.

• **DECONTAMINATION PIPELINE:** Contaminant species were detected and removed using the decontam R package.²¹ We used the most stringent hyperparameter value ($P^* = .5$) for both frequency-based and prevalence-based contaminant identification of the *isContaminant* function. DNA concentrations for the frequency method were measured by qPCR and were obtained during library preparation. The scores from the frequency and prevalence methods were combined using the “minimum” approach. The species not identified as contaminants with decontam were further excluded if their relative abundances show an inverse correlation with DNA concentration ($\rho < -0.2$, $P < .05$, Spearman's correlation).²²

• **STATISTICAL ANALYSES:** Statistical analyses were performed using R (v4.0.2) software packages. Shannon diversity indices and Bray-Curtis dissimilarity indices were respectively computed by the function *diversity* and function *vegdist* in the *vegan* R package. Two-sided Wilcoxon's rank-sum test was used to compare Shannon diversity and Bray-Curtis dissimilarity indices between 2 groups. Principal coordinates analysis (PCoA) was performed using the *dudi.pco* function in the *ade4* R package. For all boxplots, the box edges denoted the first and third quartiles and the horizontal line denoted the median, with the whiskers extending up to the 1.5-fold interquartile ranges.

The association between disease status and microbial features (species and gene families) was determined using multivariable regression implemented in MaAsLin2

(<https://huttenhower.sph.harvard.edu/maaslin>). Each microbial feature was modeled as a function of diagnosis as a categorical variable, age as a continuous covariate, and sex as a binary covariate. Informative species that contribute to the distinguishment between patients with or without oGVHD were selected using least absolute shrinkage and selection operator (LASSO) logistic regression analysis. We included species presented in at least 2 samples in LASSO modeling. We used the *glmnet* R package to build the models. The microbial features were dichotomized as present or absent. The lambda parameter was based on 5-fold cross-validation, and the value that minimized the mean cross-validated error was selected. The whole process was repeated 100 times, and the microbial features with nonzero coefficients in all models were retained.

We used logistic regression models to estimate whether the presence of a species was associated with the ICO score (6-11) and the severity of oGVHD-related clinical signs, including Schirmer's test score, CFS score (0-3), degree of conjunctival hyperemia (0-2), degree of symblepharon (0-1), degree of subtarsal fibrosis (0-2), and degree of MGL (0-6). The following characteristics were included for screening factors that potentially contribute to the presence of a species in patients with oGVHD: age, sex (recipients and donors), post allo-HSCT days, donor match, primary disease, stem cell source, donor type, match of blood type between a recipient and a donor, medical history, chemotherapy, and radiotherapy history. Univariate analysis was performed with all the characteristics. Characteristics that

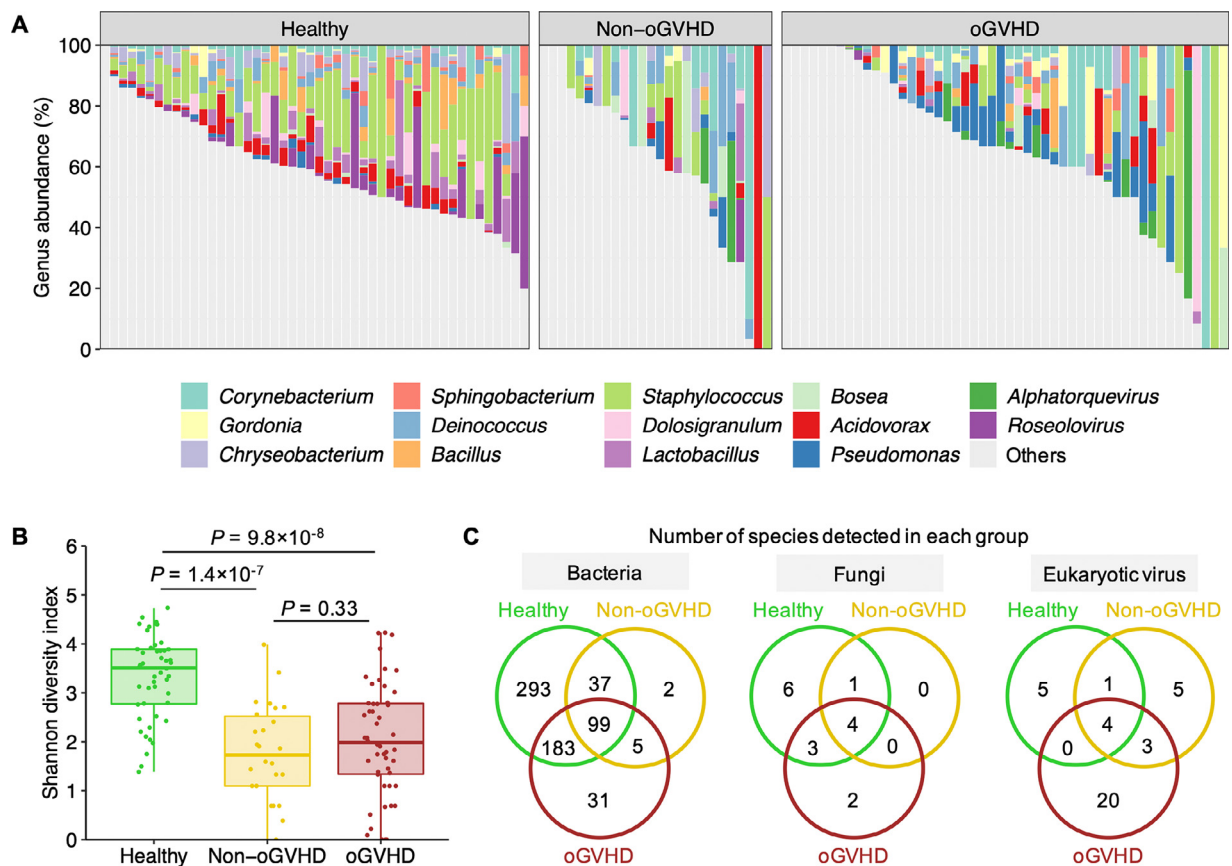


FIGURE 1. The ocular surface microbiome compositions of healthy subjects and allo-HSCT recipients with or without oGVHD. **A.** Microbiome of healthy subjects (n = 48), allo-HSCT recipients without oGVHD (non-oGVHD, n = 26), and recipients with oGVHD (n = 50) classified at the genus level. **B.** The α -diversity measured with the Shannon index for healthy subjects, non-oGVHD recipients, and patients with oGVHD. P values were computed using 2-sided Wilcoxon's rank-sum test. **C.** Number of bacterial, fungal, and eukaryotic viral species that were unique for the healthy, non-oGVHD, and oGVHD groups and shared across all groups. allo-HSCT = allogeneic hematopoietic stem cell transplantation, oGVHD = ocular graft-versus-host disease.

were significantly associated with a species ($P < .1$) were further analyzed using a multivariable logistic regression model.

RESULTS

• **OVERVIEW OF THE OCULAR SURFACE MICROBIOME IN ALLO-HSCT RECIPIENTS:** The study cohort was composed of 48 healthy subjects and 76 patients who underwent allo-HSCT, including 26 recipients without the development of oGVHD (denoted as non-oGVHD) and 50 recipients experiencing chronic oGVHD. The demographic characteristics for the study population are detailed in Table 1.

Overall, bacteria accounted for the majority of the ocular surface microbiota in both healthy individuals and patients after allo-HSCT (Supplemental Figure 1). However, the ocular surface microbiota of the allo-HSCT recipients exhibited an increased interindividual variation com-

pared with healthy individuals (Figure 1, A). This trend was quantitatively confirmed by β -diversity measured with Bray-Curtis dissimilarities (healthy vs non-oGVHD and healthy vs oGVHD, both $P < 2.2 \times 10^{-16}$). No significant difference ($P = .70$) in β -diversity was observed between the allo-HSCT patients with and without oGVHD.

The α -diversity (within-sample diversity) measured with the Shannon index was significantly lower in allo-HSCT recipients than healthy individuals (healthy vs non-oGVHD, $P = 1.4 \times 10^{-7}$; healthy vs oGVHD, $P = 9.8 \times 10^{-8}$; Figure 1, B). We did not find significant difference ($P = .33$) in α -diversity between the allo-HSCT patients with and without oGVHD. These results imply that the loss of diversity in patients with oGVHD was predominantly driven by process involved in allo-HSCT.

Consistent with a decrease in α -diversity, the majority of bacterial and fungal species present in the healthy subjects' ocular surface were absent in the patients after allo-HSCT (Figure 1, C). Although the prevalence of specific viral species was not significantly differed at the individual

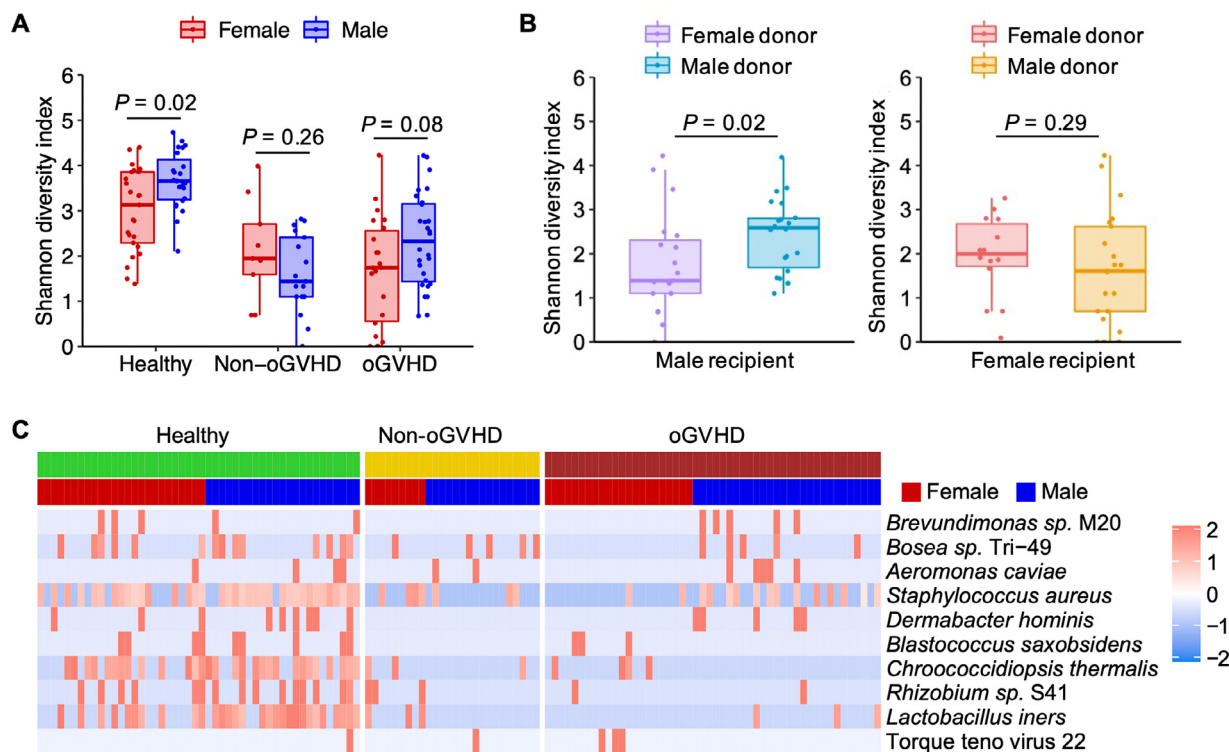


FIGURE 2. Influence of sex at the ocular surface microbiome in patients who underwent allo-HSCT. **A.** The α -diversity of male and female participants in the healthy, non-oGVHD, and oGVHD groups. **B.** The α -diversity of male and female allo-HSCT recipients with sex-consistent or sex-inconsistent donors. **C.** Species showing differences in abundance ($P < .05$) between male and female allo-HSCT recipients in the non-oGVHD or GVHD group. *Lactobacillus iners* ($P = .024$) was enriched in male individuals in the healthy group. *Lactobacillus iners* ($P = .015$) and *Rhizobium* sp. S41 ($P = .015$) were enriched in female recipients in the non-oGVHD group. *Dermabacter hominis* ($P = .041$), *Staphylococcus aureus* ($P = .007$), *Brevundimonas* sp. M20 ($P = .04$), *Bosea* sp. Tri-49 ($P = .041$), and *Aeromonas caviae* ($P = .041$) were enriched in male patients in the oGVHD group, whereas Torque teno virus 22 ($P = .049$), *Chroococcidiopsis thermalis* ($P = .021$), and *Blastococcus saxobidensis* ($P = .049$) were enriched in female patients in the oGVHD group. Relative abundances in the heatmap were centered and scaled across all the samples. allo-HSCT = allogeneic hematopoietic stem cell transplantation, oGVHD = ocular graft-versus-host disease.

level between healthy subjects and allo-HSCT recipients, we identified more types of eukaryotic viruses at the group level in samples from allo-HSCT recipients compared with those from healthy subjects. Specifically, the viruses of families *Anelloviridae* and *Papillomaviridae* were diverse in allo-HSCT recipients (Supplemental Figure 2).

• **SEX-DEPENDENT ALTERATIONS OF THE OCULAR SURFACE MICROBIOME:** We found that in healthy individuals the α -diversity of the ocular surface microbiota of males was higher than females (Figure 2, A; $P = .02$). This trend held in the allo-HSCT recipients with oGVHD ($P = .08$) but was not observed in the non-oGVHD recipients ($P = .26$). To investigate effects of the donor-recipient combination, we compared the α -diversity for male and female recipients, between those who received grafts from sex-matched and sex-mismatched donors. Interestingly, the α -diversity of male recipients with female donors was lower than that with male donors (Figure 2, B; $P = .02$). In contrast, donor sex had no significant effect on the α -diversity of female recipients ($P = .29$).

We next examined the species with differences in abundance between male and female recipients either without or with oGVHD (Figure 2, C). For patients with oGVHD, *Staphylococcus aureus* was found almost exclusively ($P = .007$) in males, whereas for the healthy group ($P = .34$) and the non-oGVHD group ($P = 0.098$), *S. aureus* showed no significant difference in abundance between males and females. In addition, *Brevundimonas* sp. M20 ($P = .04$) and *Dermabacter hominis* ($P = .04$) were enriched in male recipients with oGVHD but were rarely detected in the samples from non-oGVHD patients. These results indicate that sex has an effect on the pattern of alterations in the ocular surface microbiome of allo-HSCT recipients, which may be in turn associated with transplantation outcomes such as the development of oGVHD. Hence, we included sex as a covariate in the subsequent analysis.

• **LOSS OF DIVERSITY IN THE OCULAR SURFACE MICROBIOTA OF PATIENTS AFTER ALLO-HSCT:** Our observation of a marked loss of microbiota diversity at the ocular surface in allo-HSCT recipients both with and without oGVHD

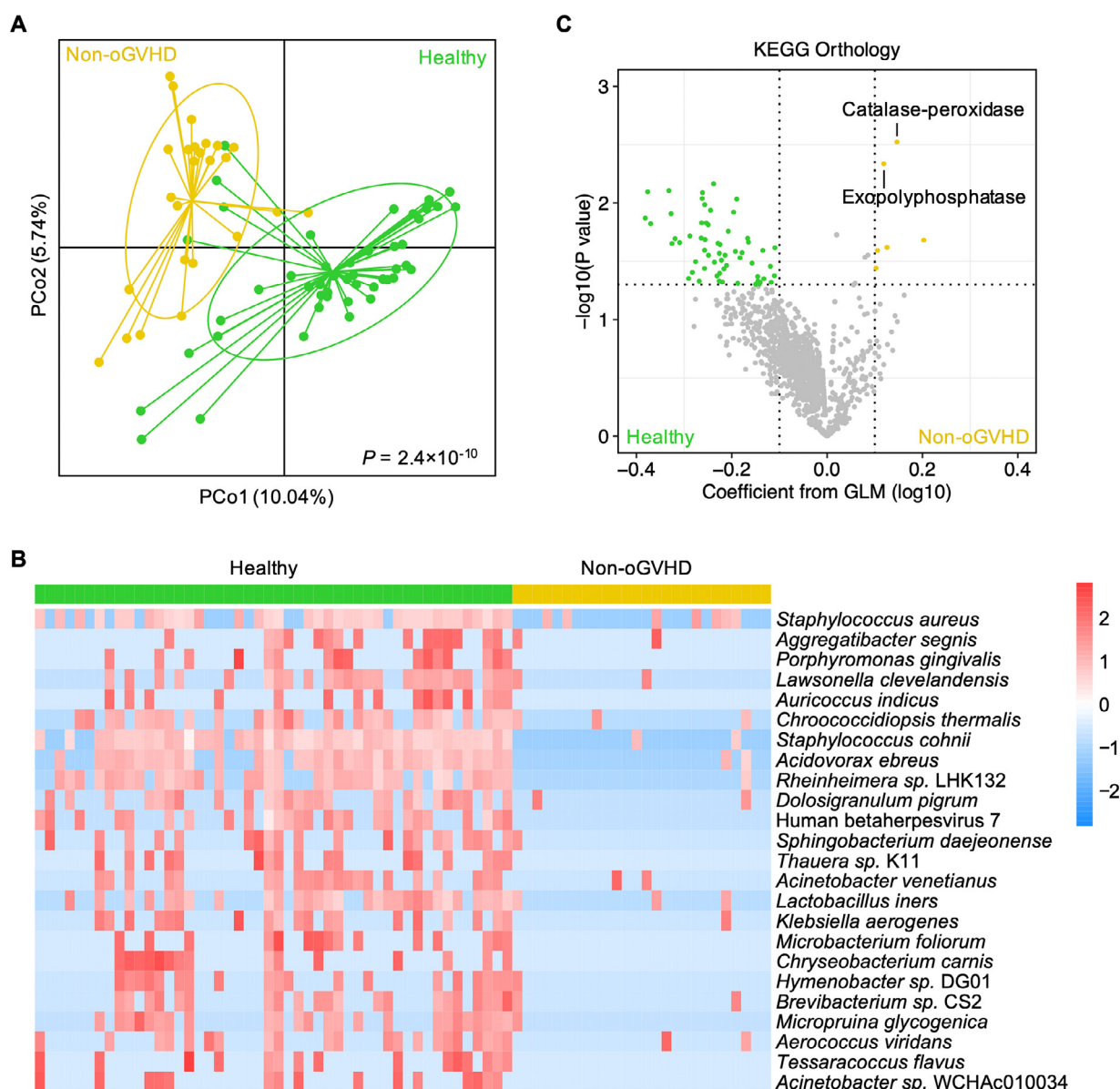


FIGURE 3. The ocular surface microbiome differences between healthy individuals ($n = 48$) and non-oGVHD patients ($n = 26$). **A.** Principal coordinates (PCo) analysis based on the Bray-Curtis distance for species-level composition of samples from healthy and non-oGVHD participants. **B.** Distribution of the commensals (present at least in 30% of the healthy individuals) that were depleted ($P < .05$) in the non-oGVHD patients. **C.** Functional differences summarized in KEGG orthology between healthy individuals and non-oGVHD patients. GLM = general linear model, KEGG = Kyoto Encyclopedia of Genes and Genomes, oGVHD = ocular graft-versus-host disease.

indicates that microbial dysbiosis presents at the ocular surface of post allo-HSCT individuals, regardless of the development of oGVHD. This phenomenon may be due to complicated processes involved in allo-HSCT (such as conditioning regimens) and other factors.

To further assess the post allo-HSCT microbial dysbiosis independent of oGVHD, we compared the ocular surface microbiome between healthy subjects and non-oGVHD recipients. The PCoA of the species composition showed a clear delineation between the healthy subjects

and non-oGVHD recipients (Figure 3, A). The delineation was due to a general depletion of species that were commonly presented at the ocular surface of healthy individuals (Figure 3, B).

We also examined the functional differences in the ocular surface microbiome between healthy subjects and non-oGVHD patients. As expected, the majority of gene families were reduced in the metagenomes from the non-GVHD patients (Figure 3, C), possibly resulting from the elimination of commensal microorganisms. Collectively, the oc-

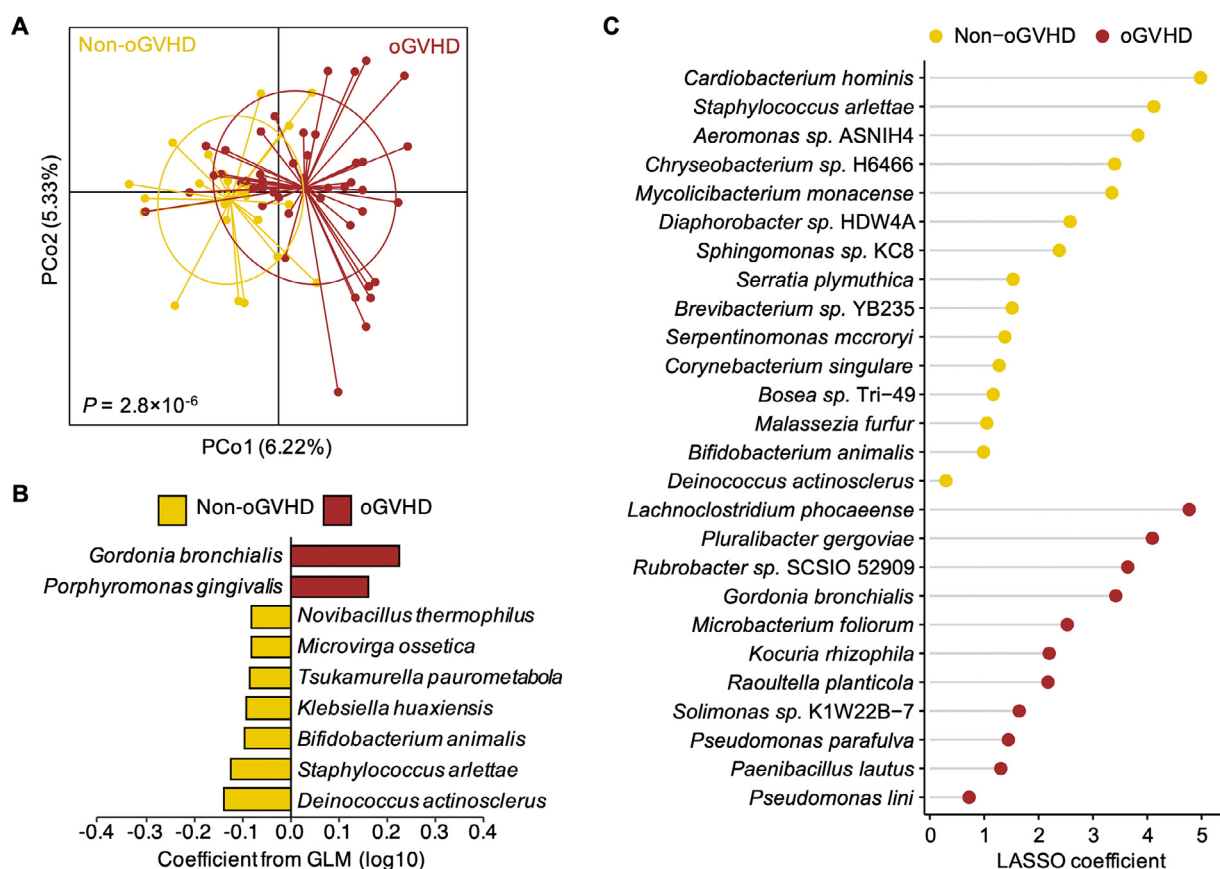


FIGURE 4. The ocular surface microbiome differences between allo-HSCT recipients without oGVHD ($n = 26$) and with oGVHD ($n = 50$). **A.** Principal coordinates (PCo) analysis of non-oGVHD and oGVHD samples based on the species-level Bray-Curtis distance. **B.** GLM coefficients of species showing differences between non-oGVHD and oGVHD patients. **C.** Mean coefficients across 100 LASSO logistic regression models for distinguishing oGVHD patients from non-oGVHD patients. Species with nonzero coefficients from 100 LASSO models were shown. allo-HSCT = allogeneic hematopoietic stem cell transplantation, GLM = general linear model, LASSO = least absolute shrinkage and selection operator, oGVHD = ocular graft-versus-host disease.

ular surface microbiota injury in patients who underwent allo-HSCT is characterized by loss of commensals and their metabolic functions.

• **THE OCULAR SURFACE MICROBIOME SIGNATURES OF OGVHD:** According to the PCoA based on the species composition, the ocular surface microbiota of oGVHD patients was distinct from that of non-oGVHD recipients (Figure 4, A). To identify species signatures of oGVHD, we compared the samples from oGVHD patients and non-oGVHD recipients using 2 independent methods. We first performed per-species association analysis using general linear models implemented in Maaslin2 and found that *Gordonia bronchialis* ($P = .04$) and *Porphyromonas gingivalis* ($P = .02$) were enriched in patients with oGVHD (Figure 4, B). In addition, several species such as *Deinococcus actinosclerus* ($P = .02$) and *Staphylococcus arlettae* ($P = .01$) were enriched in non-oGVHD recipients. We further employed LASSO logistic regression to select predictive species for oGVHD detection. Among the species with nonzero coefficients across

100 LASSO regression models (Figure 4, C), a few other species were enriched in patients with oGVHD, including *Pseudomonas parafulva*.

• **ASSOCIATIONS BETWEEN THE OCULAR SURFACE MICROBIOME AND OGVHD SEVERITY:** We next explored whether the ocular surface microbiome could partially account for the variability in oGVHD severity. The patients with oGVHD were classified into mild-moderate and severe groups according to the ICO criteria. We found that the α -diversity of the patients experiencing severe oGVHD was higher than those with mild-moderate oGVHD (Figure 5, A; $P = .04$). We further searched for species having an increased prevalence in patients with severe oGVHD (Figure 5, B). Notably, *G. bronchialis* ($P = .07$) and *P. parafulva* ($P = .04$), which were enriched in oGVHD patients compared with non-oGVHD recipients, were more prevalent in patients experiencing severe oGVHD than those with mild-moderate oGVHD. In addition, the prevalence of *Acidovorax sp. 1608163* ($P = .02$), *Staphylococcus cohnii*

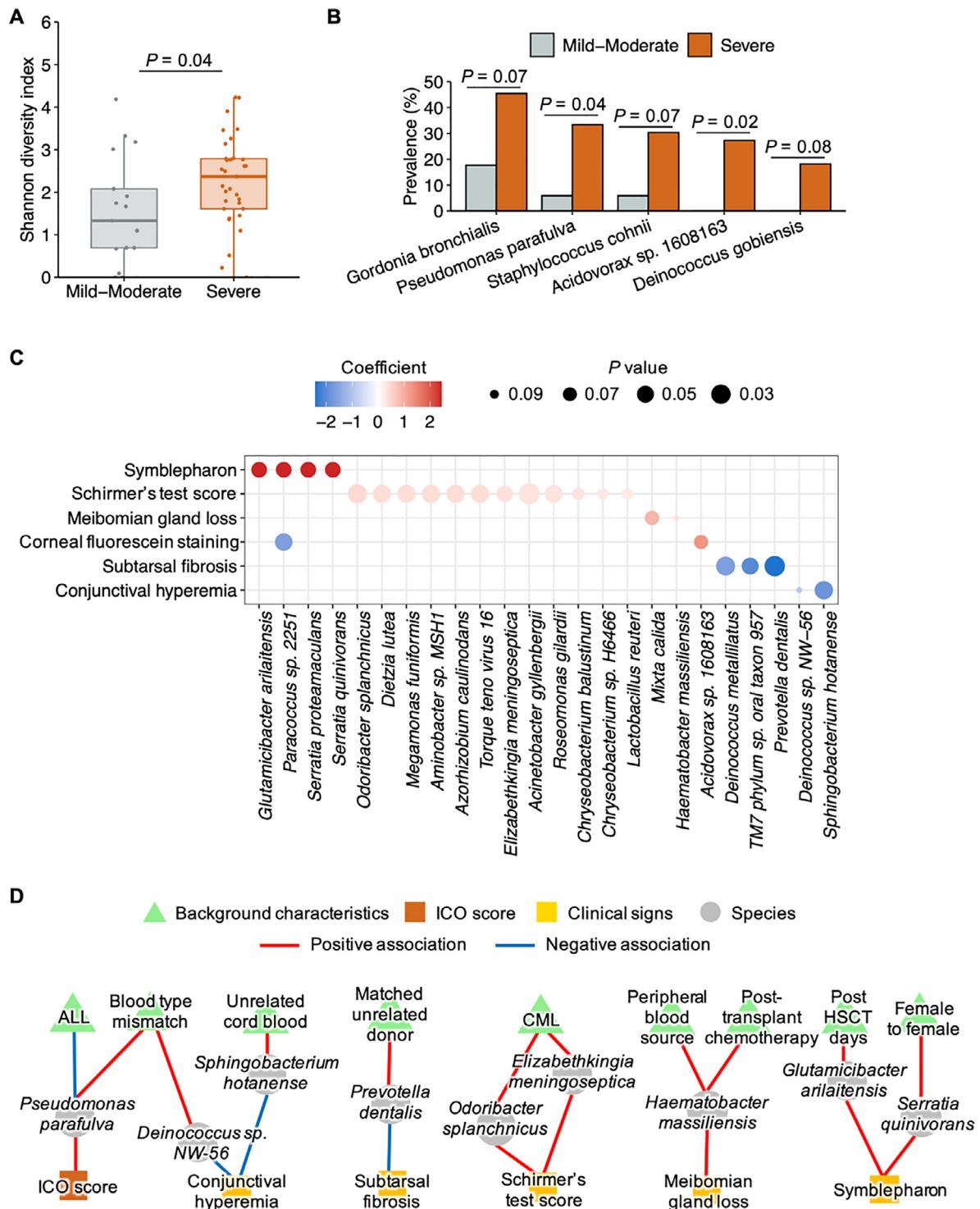


FIGURE 5. Associations of the ocular surface microbiome with oGVHD severity and clinical manifestations. A. The α -diversity of patients with mild/moderate or severe oGVHD based on the ICO score (mild/moderate: 5-8 points; severe: 9-11 points). B. Species with an increased prevalence (Fisher's exact test, $P < .1$) in patients with severe oGVHD. C. Correlations of the species presence with degree of symblepharon, Schirmer's test score, degree of meibomian gland loss, corneal fluorescein staining score, degree of subtarsal fibrosis, and degree of conjunctival hyperemia, respectively ($P < .1$). Point color indicates coefficients of logistic regression models, whereas point size indicates P values. D. Associations between the presence of species that were correlated with at least 1 clinical sign of oGVHD or the ICO score and background characteristics of oGVHD patients. ALL = acute lymphocytic leukemia, CML = chronic myeloid leukemia, HSCT = hematopoietic stem cell transplantation, ICO = International Chronic Ocular GVHD, oGVHD = ocular graft-versus-host disease.

($P = .07$), and *Deinococcus gobiensis* ($P = .08$) was slightly higher in patients with severe oGVHD than patients with mild-moderate oGVHD. These results indicate that the severity of oGVHD is associated with the ocular surface microbiome.

Apart from the overall severity of oGVHD, patients with oGVHD exhibited variation in ocular manifestations (Supplemental Table 1). Intriguingly, we observed that patients with oGVHD also showed marked individuality in the ocular surface microbiome. The fruitful demographical and clinical characteristics (Table 1 and Supplemental Table 2) enabled us to evaluate the factors associated with the interindividual variation found in patients with oGVHD. Associations between the presence or absence of microbial species and ocular manifestations are shown in Figure 5, C. We observed that degree of symblepharon was positively associated with the presence of *Serratia* species including *S. quinivorans* ($P = .06$) and *S. proteamaculans* ($P = .06$). Consistent with previous observations that many genera were depleted in patients with dry eye disease,⁸ Schirmer's test score was positively associated with a group of species. *Mixta calida* ($P = .07$) was positively associated with MGL. *Paracoccus* sp. 2251 ($P = .05$) was negatively associated with CFS score, whereas *Acidovorax* sp. 1608163 was positively associated with CFS score ($P = .07$). *Deinococcus metallilatus* ($P = .03$), *Prevotella dentalis* ($P = .02$), and TM7 phylum sp. oral taxon 957 ($P = .05$) were negatively associated with sub tarsal conjunctival fibrosis. *Sphingobacterium hotanense* ($P = .04$) was negatively associated with conjunctival hyperemia.

We further performed association analysis between the presence or absence of microbial species and patient characteristics (Figure 5, D). Notably, *P. parafulva* showed a slightly increased prevalence in allo-HSCT recipients with blood-type mismatched donors ($P = .07$). Post-transplant chemotherapy was positively correlated with the presence of a group of species (Supplemental Figure 3), including *Haematobacter massiliensis* ($P = .03$), which was weakly associated with the degree of MGL ($P = .09$). In addition, we identified several species whose presence was positively associated with days after allo-HSCT. Collectively, these observations suggest that multiple factors (such as previous courses of treatment) may have an impact at the ocular surface microbiome and thus potentially contribute to the risk of developing severe oGVHD.

DISCUSSION

Chronic ocular GVHD is a common complication of allo-HSCT and affects an increasing population of patients given the increasing number of transplant recipients.²³ Recipients of allo-HSCT are recovering from conditioning-related tissue damage, which may disrupt the mucosal barriers and homeostasis between the host and the ocular surface microbiota. Furthermore, they experience periods of lack-

ing and building a functioning immune system before and after engraftment, respectively. Such perturbations lead to a breach in immune tolerance for beneficial microbiota and failure of the immune system to fend off pathogens. It is well established in the gut that the microbiota diversity significantly decreases after allo-HSCT and the disruption of microbiota is linked to the risk of GVHD.⁶ These aspects suggest an essential role of the ocular surface microbiota in the development of oGVHD.

Using metagenomic shotgun sequencing, we characterized the microbiome injury in patients after allo-HSCT and identified microbial signatures in patients with oGVHD. We also demonstrated the links between the ocular surface microbiome and oGVHD severity. In addition, we explored the associations between potential risk factors and the variability in the ocular surface microbiome. Our study provides novel insights into the pathophysiology of oGVHD and has implications of microbiome-target preventive and therapeutic avenues of oGVHD in the future.

We observed a loss of diversity in the ocular surface microbiota driven by depletion of commensal species in patients who underwent allo-HSCT. This trend is consistent with previous observations in the gut microbiota.⁷ The post-transplant loss of diversity may be due to multiple factors in previous courses of conditioning regimens and other treatments including chemotherapy, irradiation, oral antibiotic therapy, and so on.⁶ Notably, the participants in this study did not receive topical antibiotics. It is unclear whether the local microbiota at the ocular surface is affected by oral antibiotics. The reasons why many commensal species were depleted in allo-HSCT recipients are worth investigation in future studies. We speculate that the elimination of commensal microbiota can promote the colonization of pathogenic microorganisms at the ocular surface.

A recent study demonstrated that neutrophil extracellular traps (NETs) and NET-associated proteins contribute to the pathologic changes of oGVHD at the ocular surface.²⁴ Evidence suggests that NETs are associated with microbial dysbiosis and a loss of microbiota diversity.^{25,26} We hypothesize that microbial dysbiosis of the ocular surface microbiota elicits excessive neutrophil activation. The uncontrolled or excessive formation of NETs may damage surrounding cells and thereby contribute to oGVHD pathophysiology. It is worth pursuing whether neutrophil activation serves as a pathophysiological link between microbial dysbiosis and disruption of immune homeostasis in oGVHD.

At the cohort level, we identified an increase in diversity of eukaryotic viruses at the ocular surface of allo-HSCT recipients, especially anelloviruses and papillomaviruses. Torque teno virus (TTV) was a predominant virus of a resident ocular surface virome in 65% healthy individuals²⁷ and was detected in the vitreous of patients with culture-negative postoperative endophthalmitis and seasonal hyperacute panuveitis.^{28,29} TTV activates the innate immune system via Toll-like receptor 9 signaling that generates

both serologic and T-cell-mediated responses.³⁰ Whether TTV enriched at the ocular surface contributes to the topical inflammatory response remains unclear. Previous studies demonstrated that eukaryotic viruses in the gut are increased over time after allo-HSCT.^{31,32} Moreover, persistent DNA viruses (anelloviruses, herpesviruses, papillomaviruses, and polyomaviruses) are increased in patients with enteric GVHD.³¹ In addition, immunosuppressants significantly increase the viral load in organ transplant recipients.³¹ These viruses may be potential predictors for the state of the local immune system disorder as they act in the gut. Whether they play a causal role in the development of oGVHD warrants further investigation.

Male recipients have worse outcomes compared with female recipients according to epidemiologic studies.³³ We found that healthy males and those with oGVHD exhibited higher α -diversity in the ocular surface microbiota than females, whereas non-oGVHD patients did not show this trend. Moreover, male recipients with female donors have an increased risk of chronic GVHD, whereas donor sex has no effect on outcomes for female recipients.³⁴ Interestingly, our results suggest that donor sex may influence the alteration pattern of the α -diversity for male recipients but not for female recipients. Our study preliminarily explored the difference in the ocular surface microbiota between recipients with sex-matched and sex-mismatched donors. This issue has not been investigated in previous studies. There was no formal null hypothesis to test, so that no sample size calculation was conducted *a priori*. We recognized that the sample size may not support the conclusion of whether donor-recipient combinations affect the ocular surface microbiome. Further studies based on a larger cohort are needed to answer this question. Nonetheless, our finding provides novel insights into the relationship between sex and oGVHD as well as clues for sample size calculation in future studies.

We observed that *P. parafulva* showed an increased prevalence in patients with severe oGVHD and ABO blood group mismatch. The incidence of extensive chronic GVHD is higher in ABO-mismatch patients than ABO-match patients.³⁵ Interestingly, the *FUT2* gene, which influences the risk of GVHD,³⁶ regulates ABO blood group antigens in the mucous layer and plays a role in shaping host microbiota.^{37,38} If *P. parafulva* is a modulator of this process, therapies targeting this species may be helpful for oGVHD treatment.

A culture-based study by Shimizu and associates⁹ reported that the ocular surface bacteria were more diverse in oGVHD patients than non-oGVHD recipients and healthy individuals. However, Andersson and associates⁸ performed 16S rRNA sequencing and showed that the microbial diversity was lower and numerous genera were less abundant in oGVHD patients than healthy individuals. The conflicting results may be due to the use of different detection techniques. In agreement with Andersson and associates, our study using metagenomic shotgun sequenc-

ing suggested that the diversity was generally decreased in allo-HSCT recipients. We further found that while still lower than healthy individuals, the α -diversity was higher in patients with severe oGVHD than those with mild-to-moderate oGVHD. This implies the emergence of pathobionts at the ocular surface of patients with oGVHD but not in a way of mono-domination as in the gut. The distinct features of the ocular surface microbiota may be due to the continuous exposure to the external environment. Thus, the presence of a species may be more important than its abundance at the ocular surface.

We identified multiple factors that may contribute to the increased individuality in the compositions of the microbiome in oGVHD patients, such as primary diseases and previous treatment regimens. In addition, the variation in microbiome was in turn associated with the severity of oGVHD and ocular manifestations. These associations between specific microbial species and potential risk factors for oGVHD need to be confirmed in a large cohort. Future studies revealing the mediator roles of microbiome will help develop preventive and therapeutic approaches for oGVHD. On the other hand, alterations of the ocular surface microbiota provide novel aspects for the impact of the risk factors that are poorly understood, such as use of female donor, matched unrelated donor, peripheral blood as stem cell source, and chemotherapy.

Thus far, the best practice for sampling the ocular surface microbiota has not been settled. Katzka and associates³⁹ reported that in contrast to cotton-tipped applicator and Weck-Cel cellulose sponge, calcium alginate swab was the closest to corneal epithelial biopsy. Conjunctival impression cytology has also been adopted by other studies in the ocular surface microbiota.^{40,41} However, little is known about the difference between swab and impression cytology in sampling the ocular surface microbiota. It is necessary to compare the performance of different sampling methods in the future.

Our study has several limitations. First, the sample size is relatively small. Secondly, because of the nature of the cross-sectional study, it is unknown when the microbiome dysbiosis in allo-HSCT recipients occurs and whether changes in the microbiome precede or follow the development of oGVHD. This warrants longitudinal studies to clarify the temporal dynamics throughout the transplant period. Knowing how the microbiota changes before the onset of pathology is helpful in determining whether the microbial signatures are drivers or passengers in oGVHD. Thirdly, although we used negative controls with unused microfiltration membranes, anesthetic drops were not added to the membranes. The anesthetic drops may be a source of contaminants in low-biomass material. Nonetheless, anesthetic drops were used in all participants, and our major goal was to investigate the differences in the ocular surface microbiome among healthy individuals and patients with and without oGVHD after allo-HSCT. Therefore, this issue may have little effect on the results of comparison be-

tween groups. In addition, the Alcaine eyedrops were sterilized, and each used bottle was discarded at the end of each day. Finally, the activities of the microbiota were not studied. Future studies in metatranscriptome and metabolome will help to characterize activities of the ocular surface microbiome and the host-microbiota interactions during oGVHD.

In conclusion, our study revealed the ocular surface microbial dysbiosis after allo-HSCT and investigated the microbial signatures in chronic ocular GVHD. Future efforts are required to provide a mechanistic understanding of how the ocular surface microbiome is involved in the pathogenesis of chronic ocular GVHD and other complications after allo-HSCT.

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